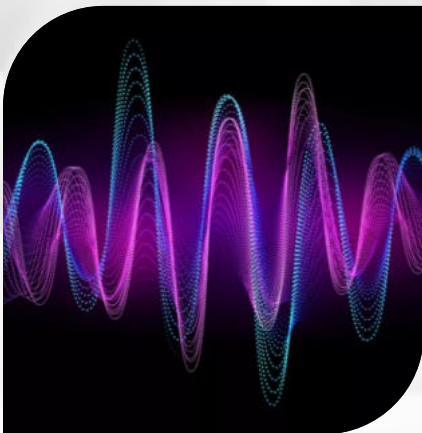
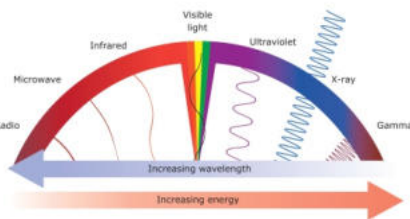


UNIT 12

ELECTROMAGNETIC SPECTRUM



DISPERSION OF LIGHT

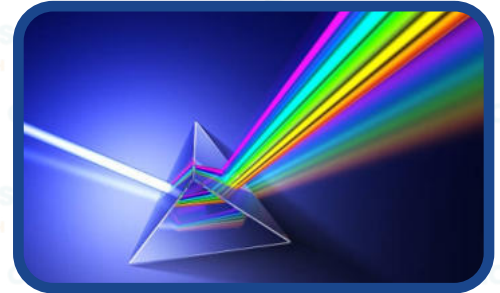
The phenomenon where light separates into different colors due to varying speeds of refraction, creating a rainbow-like effect.

SPECTRUM OF LIGHT

The entire range of electromagnetic wavelengths emitted or absorbed by different objects, encompassing visible light as well as non-visible forms like infrared and ultraviolet.

DISPERSION OF LIGHT THROUGH WATER DROPLETS

Dispersion of light through water droplets refers to the phenomenon where sunlight is refracted, reflected, and internally reflected inside individual water droplets, resulting in the separation of white light into its constituent colors, forming a rainbow.

**SPEED OF ELECTROMAGNETIC WAVES**

The speed of electromagnetic waves, denoted by 'c', is the speed at which electromagnetic radiation propagates in a vacuum, such as space. Its value is approximately 299,792,458 meters per second (m/s). The formula for the speed of electromagnetic waves is:

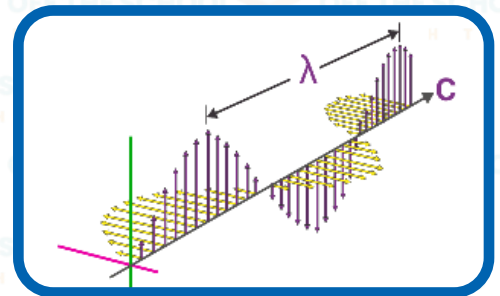
$$c = \lambda \times f$$

Where:

c = speed of light in a vacuum (m/s)

λ = wavelength of the electromagnetic wave (m)

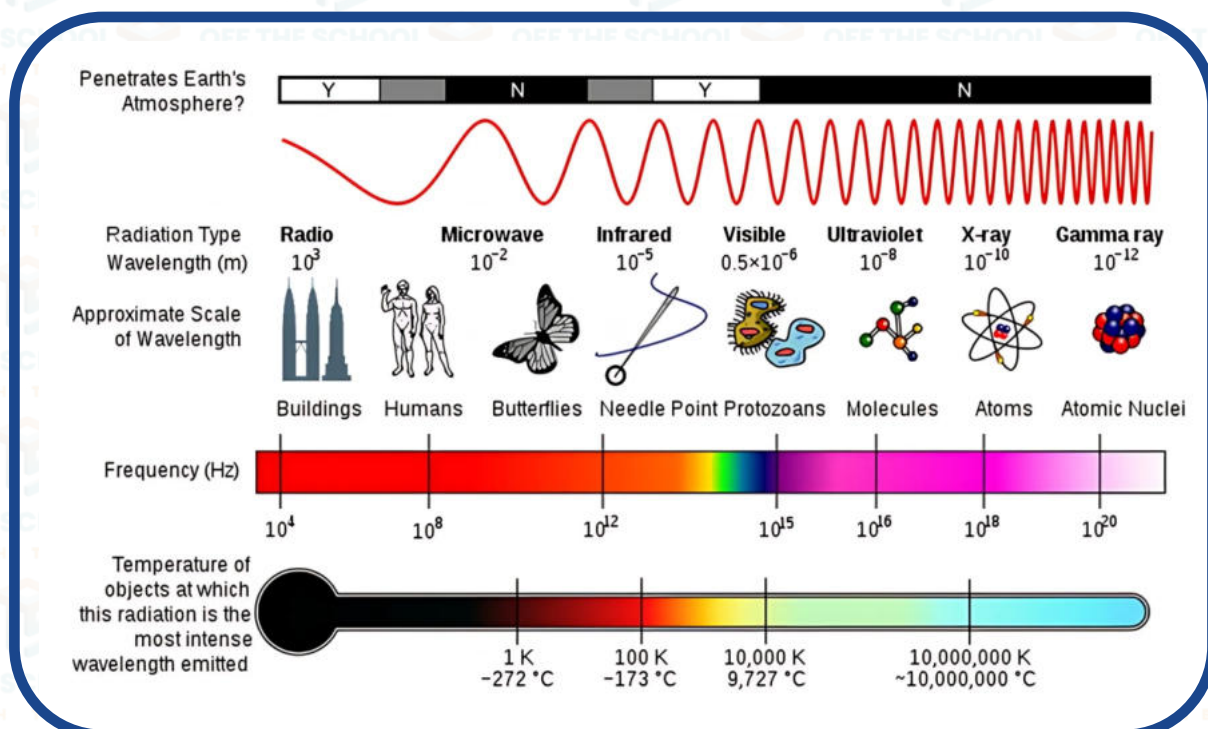
f = frequency of the electromagnetic wave (Hz)

**CHARACTERISTICS OF ELECTROMAGNETIC WAVES**

1. The oscillations of electric and magnetic fields are perpendicular to the direction of wave propagation.
2. Electromagnetic waves, including visible light, travel at a constant speed of approximately 299,792,458 meters per second (m/s) in a vacuum.
3. Electromagnetic waves have a direct relationship between their wavelength (λ) and frequency (f): $c = \lambda \times f$ where 'c' is the speed of light.
4. Electromagnetic waves cover a wide range of wavelengths and frequencies, including radio waves, microwaves, infrared, visible light, ultraviolet, X-rays, and gamma rays.
5. Electromagnetic waves can bounce off surfaces at the same angle of incidence as the angle of reflection.
6. Electromagnetic waves change direction and speed when they pass through different media with varying refractive indices.

ELECTROMAGNETIC SPECTRUM

ELECTROMAGNETIC WAVE	WAVELENGTH RANGE	FREQUENCY RANGE	SOURCES	APPLICATIONS
Radio Waves	>1 mm – km	<300 GHz	Broadcasting antennas	Radio broadcasting, telecommunications
Microwaves	1 mm – 1 m	300 Mhz– 300 GHz	Electronic devices, sun	Microwave ovens, radar, satellite communication
Infrared Waves	700 nm – 1 mm	300 Ghz – 430 THz	Hot objects, electronic devices	Remote controls, thermal imaging, infrared heating
Visible Light	400 nm – 700 nm	430 THz – 750 THz	Natural and artificial light sources	Human vision, optical communication, photography
Ultraviolet Waves	10 nm – 400 nm	750 THz – 30 PHz	Sun, specialized lamps	UV sterilization, fluorescent lamps, tanning
X-rays	0.01 nm – 10 nm	30 PHz – 30 EHZ	X-ray tubes, cosmic sources	Medical imaging, security screening, material analysis
Gamma Rays	<0.01 nm	>30 EHZ	Nuclear processes, space	Cancer treatment, nuclear medicine, astronomy



USES OF ELECTROMAGNETIC WAVES

The phenomenon where light separates into different colors due to varying speeds of refraction, creating a rainbow-like effect.

1. Communication: Radio waves and microwaves are used for broadcasting radio and television signals, as well as for wireless communication such as Wi-Fi, cellular networks, and satellite communication.

2. Microwave Heating: Microwaves are employed in household microwave ovens to heat food quickly and efficiently.

3. Infrared Heating: Infrared waves are used for heating purposes in devices like infrared heaters and saunas.

4. Lighting: Visible light is used for general illumination in homes, offices, and public spaces through various light sources like incandescent bulbs, fluorescent lamps, and light-emitting diodes (LEDs).

5. Medical Imaging: X-rays are used in radiography to visualize internal structures in medicine and dentistry. Gamma rays are used in nuclear medicine for diagnostic imaging and cancer treatment.

APPLICATIONS OF X-RAYS IN MEDICINE

1. Computed Tomography (CT) Scanning: CT scans use X-rays to create detailed cross-sectional images of the inside of the body. These images help in diagnosing various medical conditions, such as identifying injuries, tumors, or abnormalities in organs and tissues.

2. Radiation Therapy: X-rays are used in radiation therapy to treat cancer. High-energy X-ray beams are directed at cancerous cells to damage their DNA and prevent them from multiplying, effectively targeting and killing cancer cells while minimizing damage to healthy surrounding tissue.

UV WAVES APPLICATIONS

Ultraviolet (UV) waves have various important applications, including:

1. Sunbeds: UV radiation is used in tanning beds to artificially tan the skin.

2. Fluorescent Tubes: Fluorescent lamps and tubes use UV radiation to excite phosphor coatings inside the tubes, producing visible light.

3. Sterilization: UV light is utilized for germicidal purposes, disinfecting air, water, and surfaces in hospitals, laboratories, and water treatment facilities.

APPLICATIONS OF GAMMA RAYS IN MEDICINE

1. Positron Emission Tomography (PET) Imaging: Gamma rays are used to create 3D images of metabolic processes in the body, aiding in the diagnosis of various conditions.

2. Gamma Knife Radiosurgery: Precisely targeted gamma rays treat brain tumors and abnormalities without traditional surgery, minimizing risks and offering faster recovery.